

datalex



Work with data warehouses



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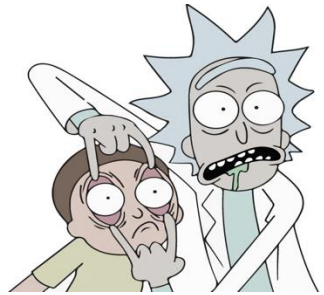
Don’t get lost



Practice



Demo



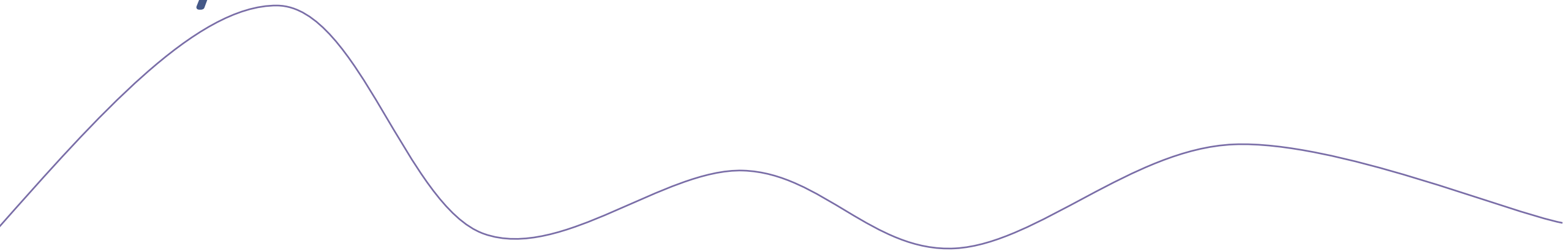
Important

Learning Objectives

1. Understand OLAP / Fact / Dimension Tables

2. Analyze data in a relational data warehouse

Analyze data in a relational data warehouse



What's Business Intelligence ?



"The term **Business Intelligence (BI)** refers to **technologies**, **applications** and **practices** for the collection, integration, analysis, and presentation of business information. The purpose of Business Intelligence is to support better business decision making. Essentially, Business Intelligence systems are data-driven **Decision Support Systems (DSS).** Business Intelligence is sometimes used interchangeably with briefing books, report and query tools and executive information systems."

OLAP = Online Analytical Processing, aka *reporting* is generally performed on databases that have been *de-normalized* to facilitate OLAP as opposed to OLTP. These databases are often referred to as *data warehouses*.

Characteristics

A **data warehouse** is a subject-oriented, integrated, time-variant and non-volatile collection of data in support of management's decision-making process.

Subject-Oriented: A data warehouse can be used to analyze a particular subject area. For example, "sales" can be a particular subject.

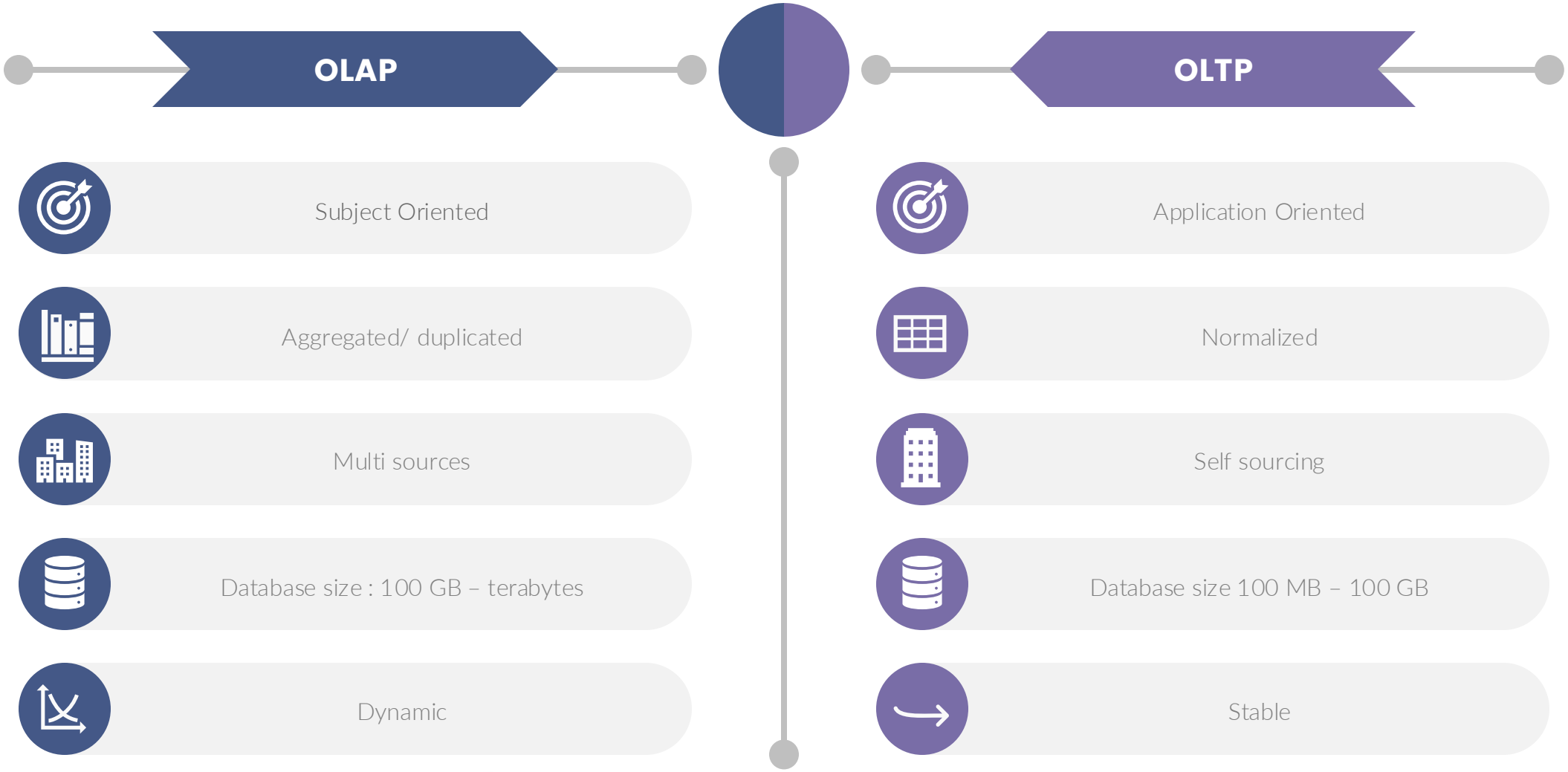
Integrated: A data warehouse integrates data from multiple data sources. For example, source A and source B may have different ways of identifying a product, but in a data warehouse, there will be only a single way of identifying a product.

Time-Variant: Historical data is kept in a data warehouse. For example, one can retrieve data from 3 months, 6 months, 12 months, or even older data from a data warehouse. This contrasts with a transactions system, where often only the most recent data is kept. For example, a transaction system may hold the most recent address of a customer, where a data warehouse can hold all addresses associated with a customer.

Non-volatile: Once data is in the data warehouse, it will not change. So, historical data in a data warehouse should never be altered.

http://olap.com/learn-bi-olap/olap-bi-definitions/business-intelligence

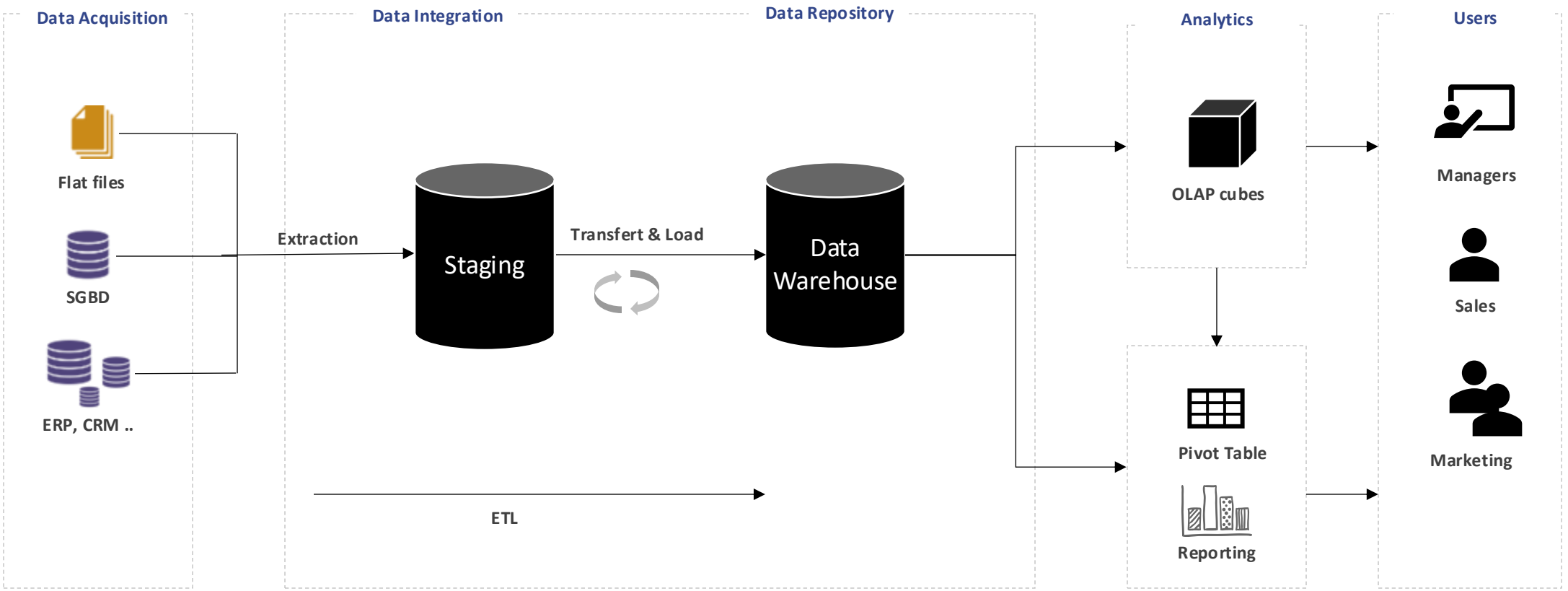
Business intelligence vs Transactional systems
Data implementation



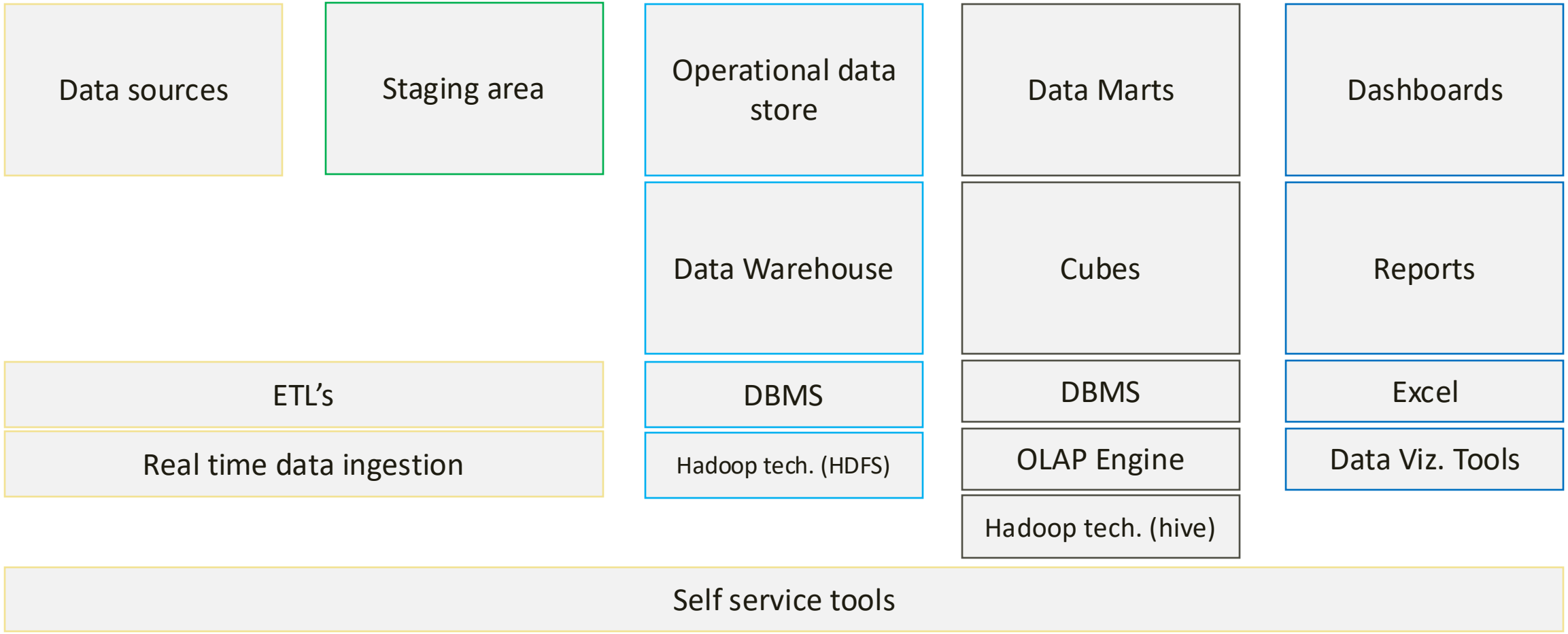
Business intelligence vs Transactional systems
Data implementation

OLAP	OLTP
Designed for analyzing the business	Designed for running the business
Summarize and refined	Detailed data
History to Current (Snapshot data)	Current up to date
Knowledge user (Manager)	Clerical User
Mostly Read (Batch Update)	Read/Update Access
Hundrend of users	Thousands of users

Architecture



The typical BI Process



Data sources

The data companies analyze through business intelligence comes from a diverse type of data sources. The most common of these are:

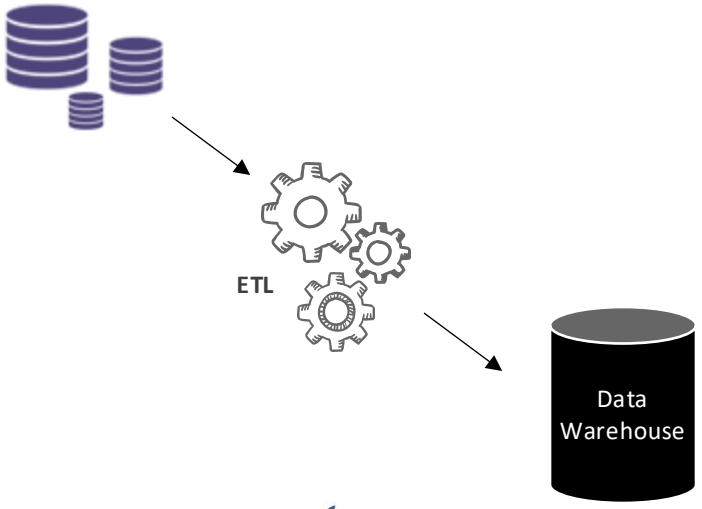
- Databases
- Flat files
- Parquet, Delta
- Web services
- Other sources such as RSS feeds

ETL

The ETL (Extraction, Transformation, Loading) process typically takes the longest to develop, and this can easily take up to 50% of the data warehouse implementation cycle or longer. The reason for this is that it takes time to get the source data, understand the necessary columns, understand the business rules, and understand the logical and physical data models.

"You get the data out of its original source location (E), you do something to it (T), and then you load it (L) into a final set of tables for the business users to query."

(Kimball's group)



Staging area

A **staging area**, or **landing zone**, is an intermediate storage area used for data processing during the extract, transform and load (ETL) process. The data staging area sits between the data source(s) and the data target(s), which are often data warehouses, data marts, or other data repositories.

Staging area is sometimes the exact copy of all integrated data.

Operational Data Store (ODS)

An **operational data store** (or "**ODS**") is a database designed to integrate data from multiple sources for additional operations on the data. Unlike a master data store, the data is not passed back to operational systems. It may be passed for further operations and to the data warehouse for reporting.

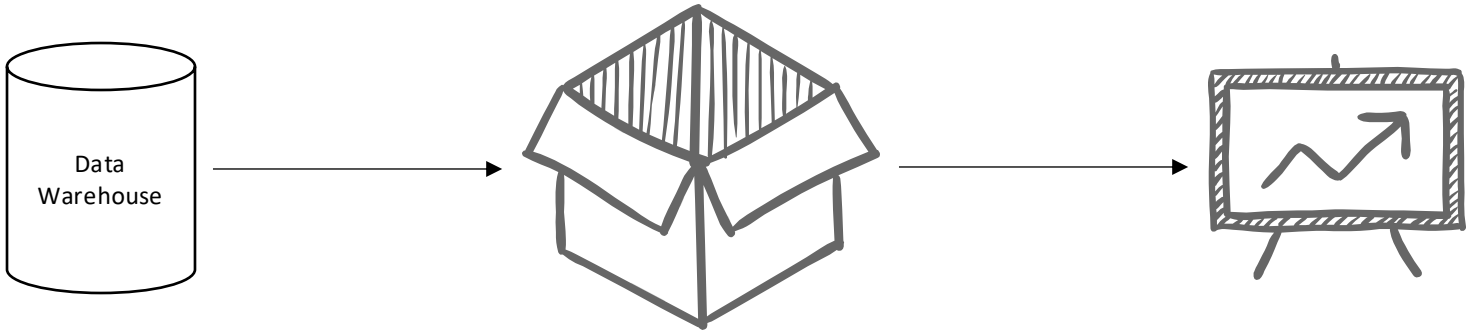
Data Warehouse



"A **data warehouse** is a subject-oriented, integrated, time-variant and non-volatile collection of data in support of management's decision making process." (Bill Inmon)

OLAP Analysis

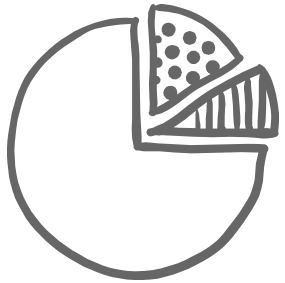
On**l**ine **A**nalytical **P**rocessing is a category of software tools that provides analysis of data stored in a database. OLAP performs multidimensional analysis of business data and provides the capability for complex calculations, trend analysis, and sophisticated data modeling



Reporting

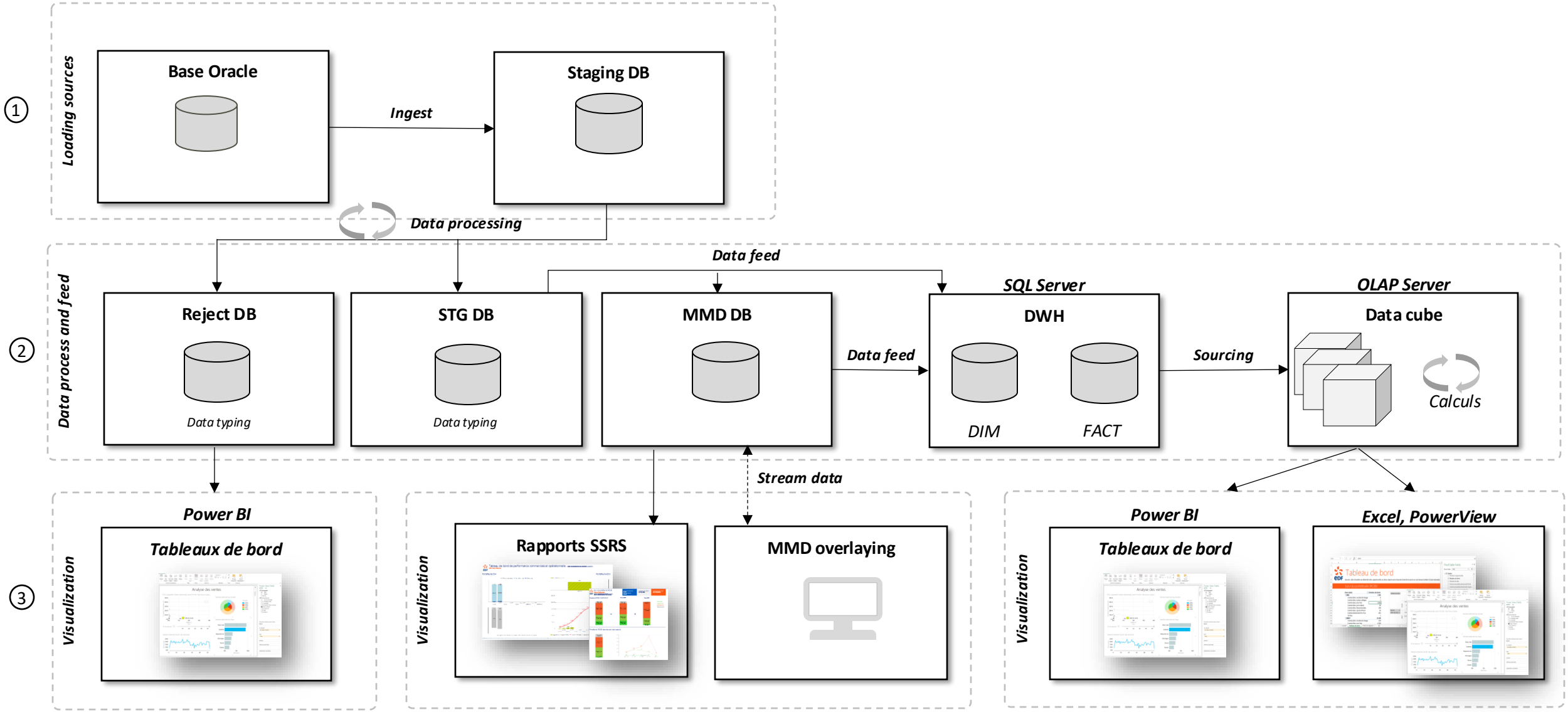
- Making reports for the client's needs.
- Presenting data and information for the end-user.
- Could be for operational or decisional use.
- Could be Static or dynamical.







Reporting

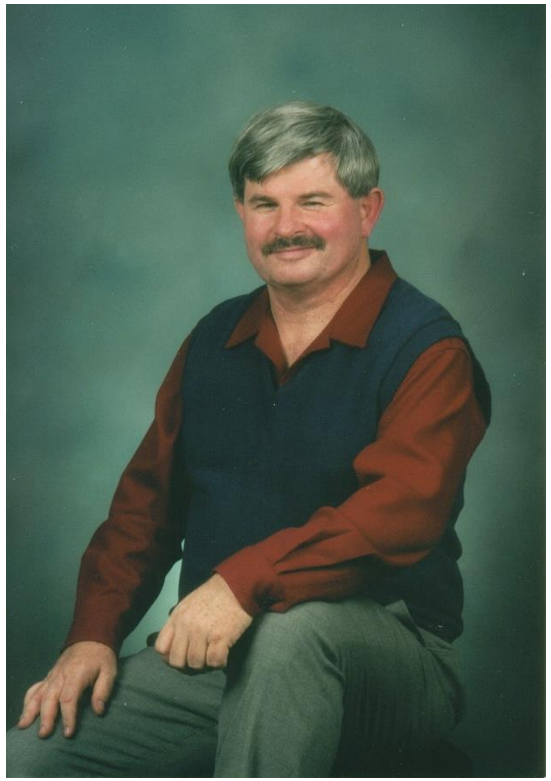

1
Loading sources
Base Oracle
Ingest
Staging DB
Data processing
2
Data process and feed
Data feed
Reject DB
Data typing
STG DB
Data typing
MMD DB
Data feed
SQL Server
DWH
DIM
FACT
Sourcing
OLAP Server
Data cube
Calculs
Stream data
3
Visualization
Power BI
Tableaux de bord
Visualization
Rapports SSRS
MMD overlaying
Visualization
Power BI
Tableaux de bord
Excel, PowerView

Two pillars

Ralph Kimball & the Kimball Group



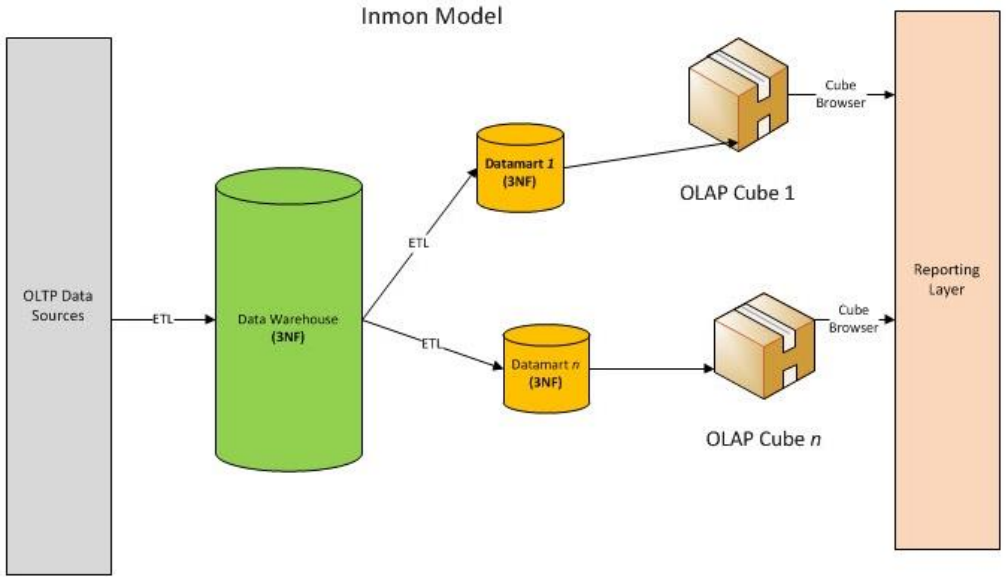
Bill Inmon



Data Mart

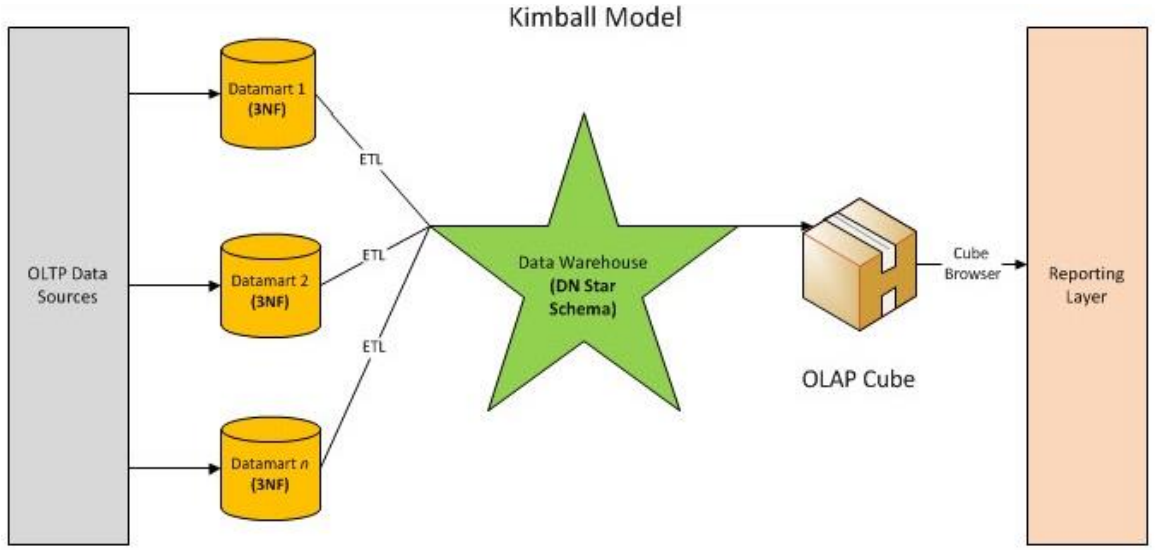
A ***Data Mart*** is conceived to meet the particular demands of a specific group of users within the organization (finance, HR, marketing...).

Inmon : bottom-up design



Leads by Data

Kimball : top-down design



Leads by Needs

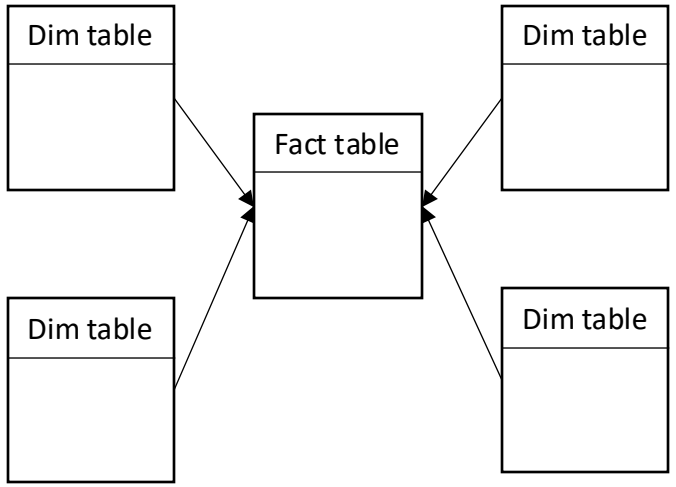
Data Mart

"You can catch all the minnows in the ocean and stack them together and they still do not make a whale." - Bill Inmon

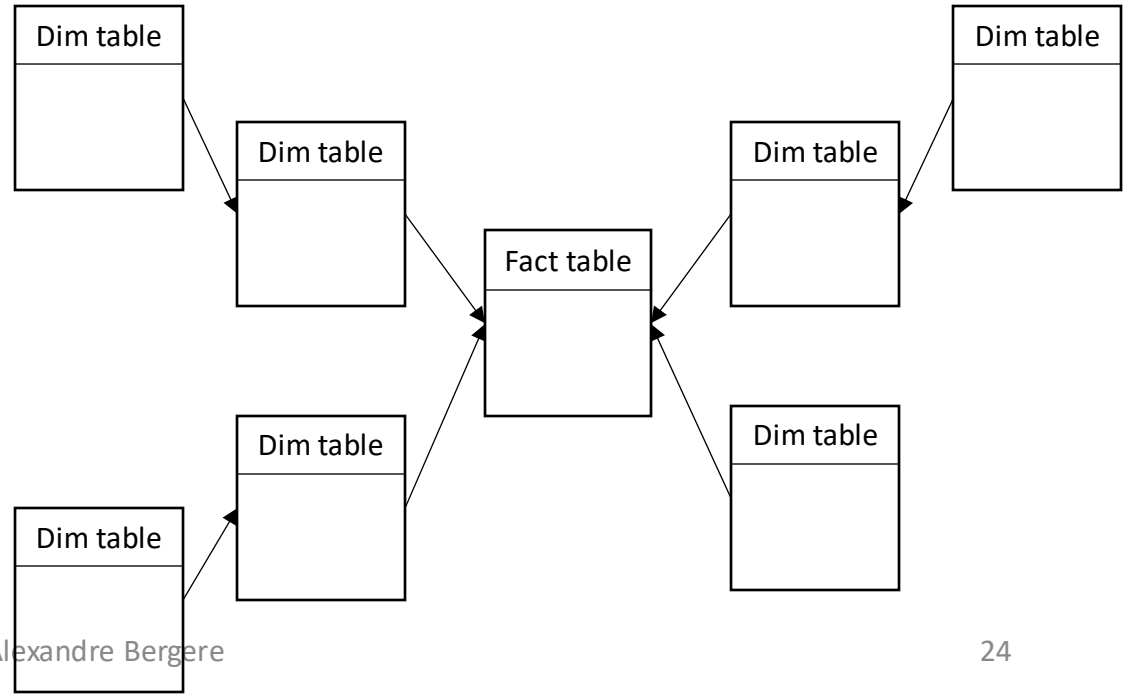
“The data warehouse is nothing more than the union of all the data marts" - Ralph Kimball

Design a data warehouse schema

Star schema : this is the model in which we have a unique fact table which is related to all other dimensions.

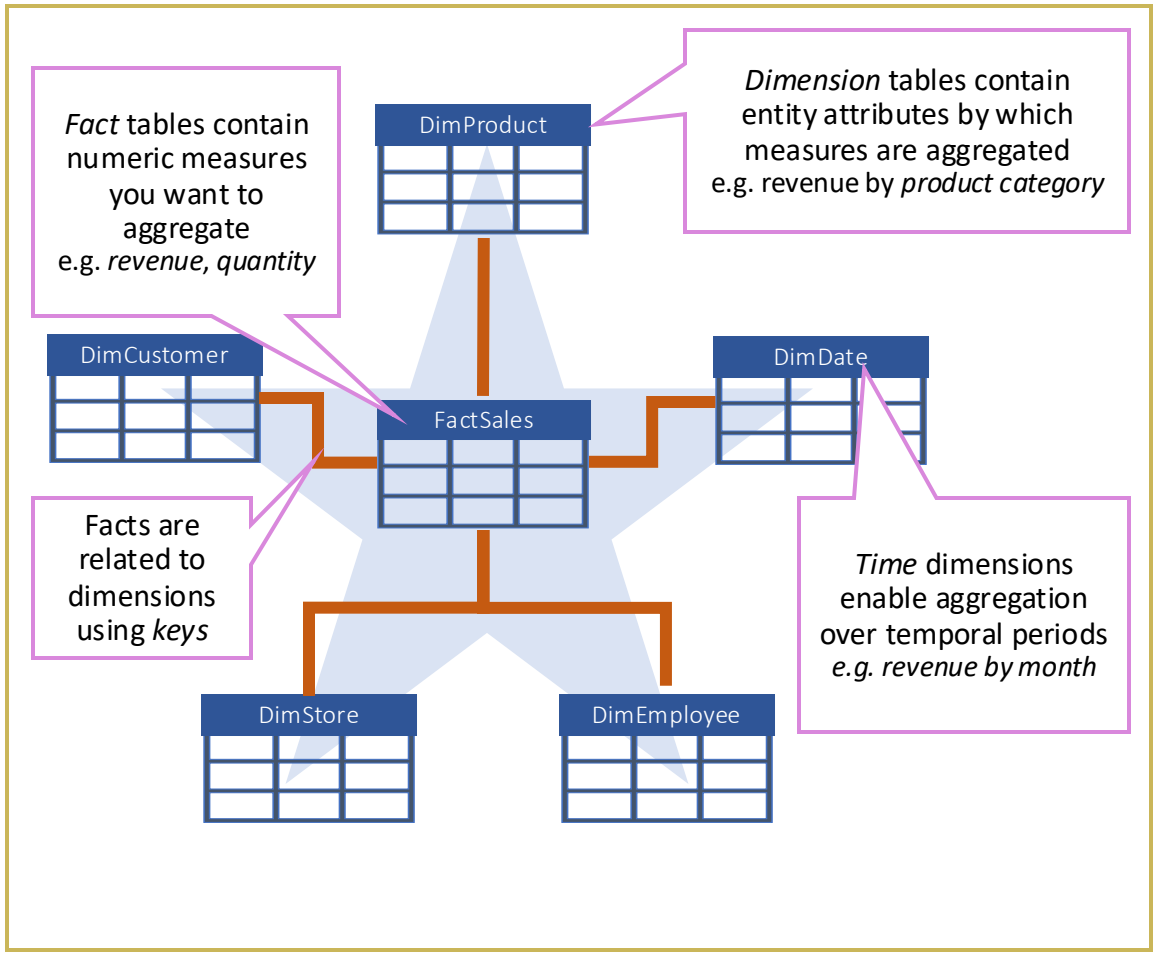


Snowflake Schema : In this type of model, dimension tables branch out and are related to other tables which have different granularities. Dimensional table are normalized into multiple lookup tables, each representing a level in the dimensional hierarchy.

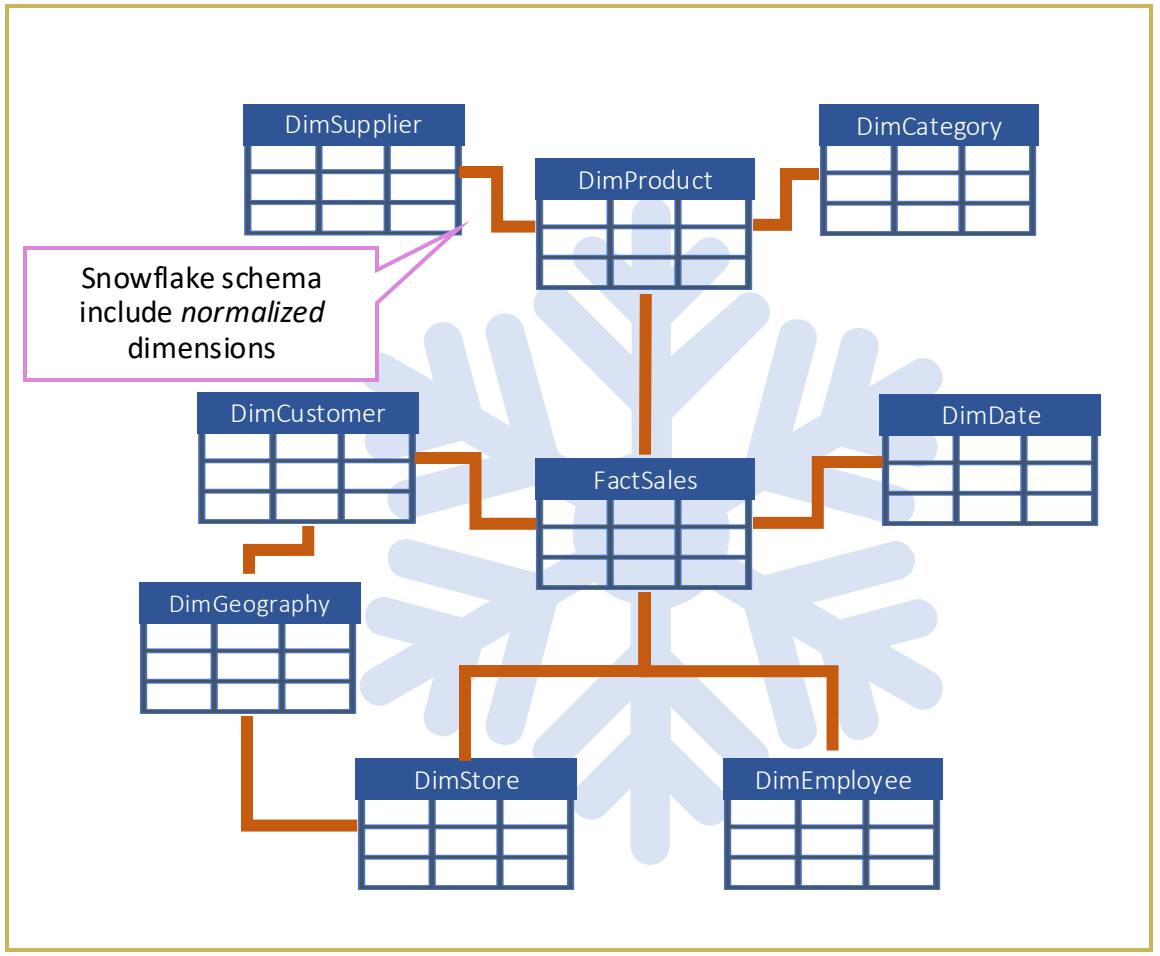


Design a data warehouse schema

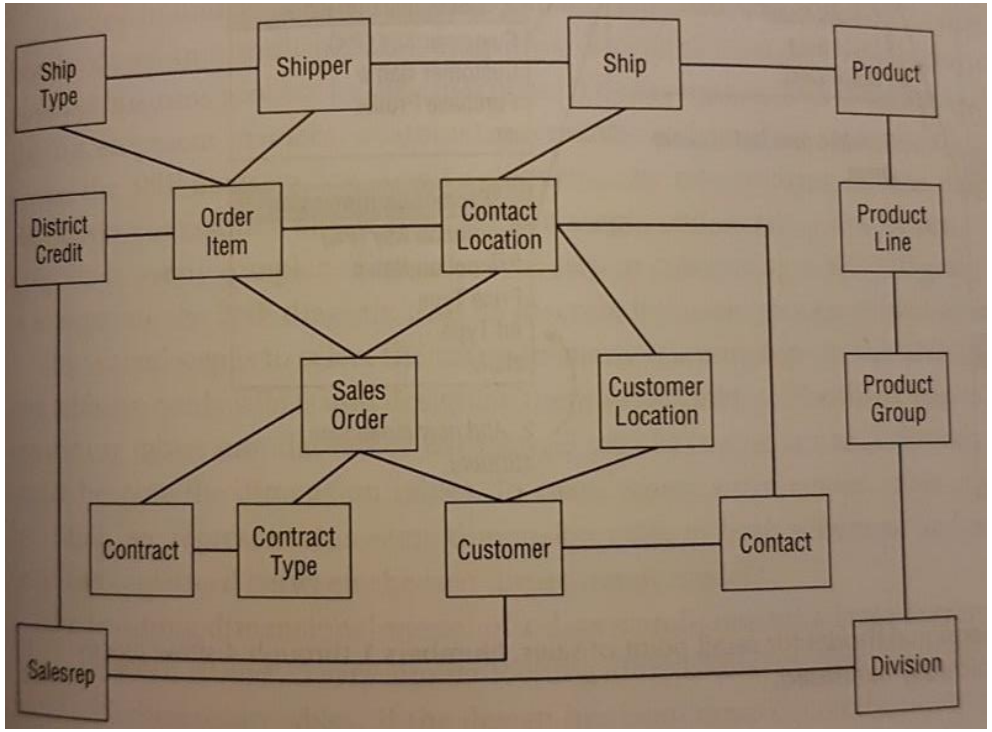
Star schema



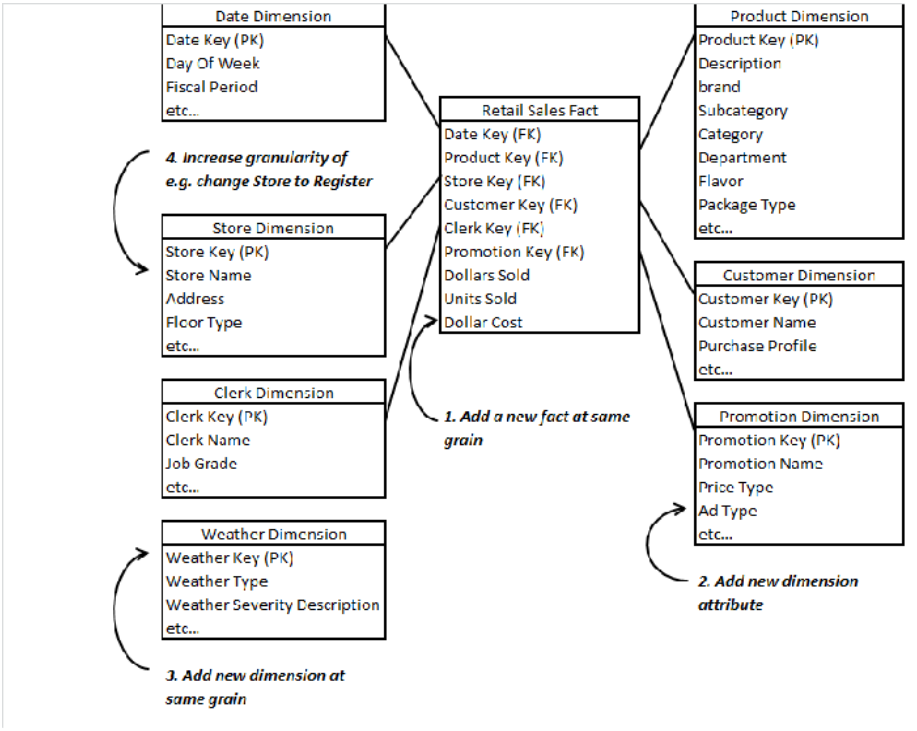
Snowflake schema



Design a data warehouse schema



3FN



DM

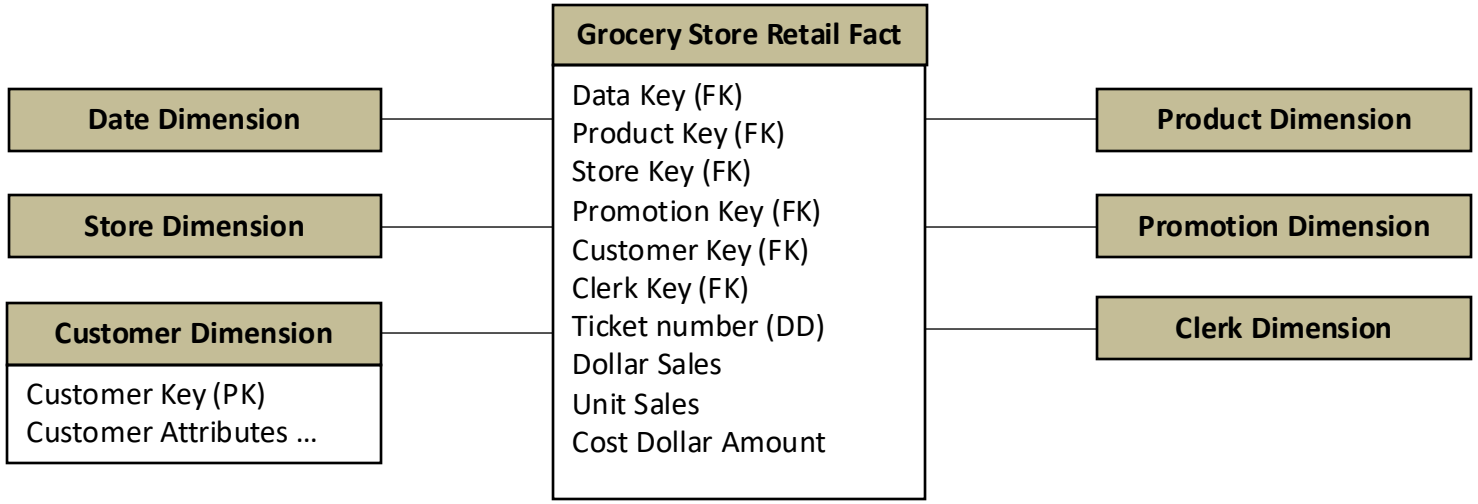
Design a data warehouse schema

	Relational table	Star schema
Advantages	• Save most storage of all modelling techniques • Simple to understand	• Simple design • Fast queries • Fits very well to OLAP models • Many DBMS are optimized for queries on star schemas
Inconvenients	• Bad performance with multiple joins • Difficult to load (ETL) • Changes are very difficult to handle	• Usually centralized on one fact table • Higher storage usage due to denormalization

Dimensions & Fact tables

Fact table : a fact table is a set of structured data that contains *measures* we would like to calculate. All attributes are of type numeric. And it contains the primary keys of the dimension tables to which it is related (surrogate key) : measurement.

Dimension table : a dimension table is an axis of analysis for the fact table. It contains attributes of different types about measurable events which are facts. It has a primary key column that uniquely identifies each dimension. Dimension tables are related to the fact table : context.





Lookups



Lookup table : a simple table with ad-hoc data and common keys to join on.



Drilling Down / Rolling Up

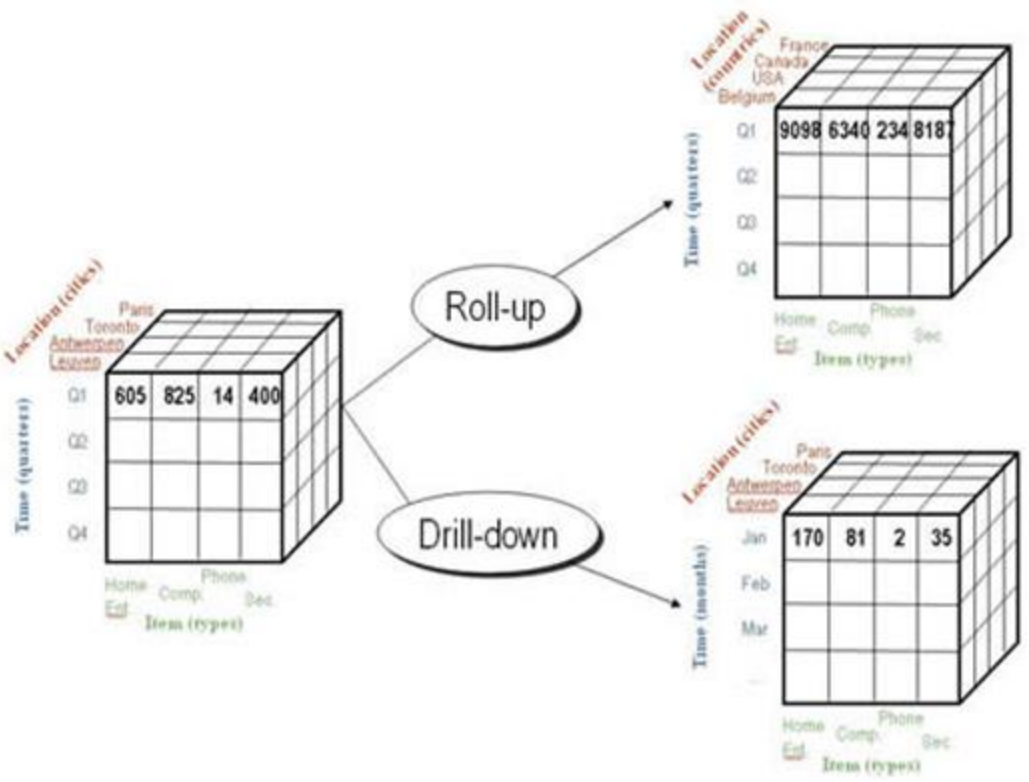
Drilling Down

“Give me more detail”

Access data which is in a lower level of a hierarchically structured database.

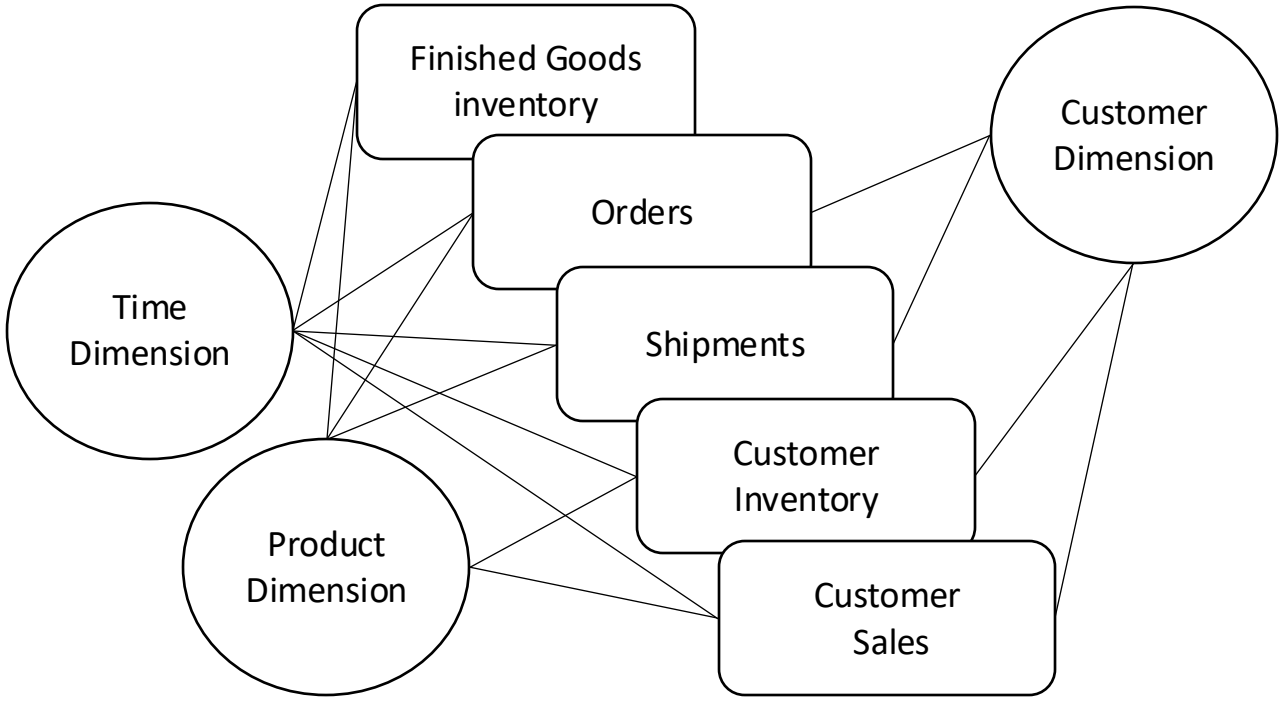
Rolling Up

“The Aggregate Navigator”



Drilling Across

- With multiples fact table:
- The new fact table called for in the drill-across operation must share certain dimensions with the fact table in the original query.
- All fact tables in a drill-across query must use conformed dimensions.
- The actual drill-across query consists of a multi-pass set of separate requests to the target fact tables followed by a simple sort/merge on the identical row headers returned from each request.



Fact Table Core Concepts

Granularity

Declaring the grain means saying exactly what a fact table record represents. A fact table record capture a measurement event.

A fact table is usually designed at a low level of Granularity. This means that we need to find the lowest level of information that can store in a fact table.

The grain ultimately is equivalent to the primary key, it’s a mistake to list a set of dimensions and the assume that this list is the grain declaration.

The grain is first anchored to a clear business process and a set of business rules.

Fact Table Core Concepts

Measure

Measures might be :

- Additive : additive measures can be summed across any of the dimensions associated with the fact table.
- Semi-additive measures can be summed across some dimensions, but not all; balance amounts are common semi-additive facts because they are additive across all dimensions except time.
- non-additive, such as ratios.

Factless table :Although most measurement events capture numerical results, it is possible that the event merely **records** a set of dimensional entities coming together at a moment in time. For example, customer request for a catalog of an investment bank.

Fact Table / Dimension Core Concepts

Surrogate key



A surrogate key is an artificial or synthetic key that is used as a substitute for a natural key.



Every join between dimension tables and facts table in a data warehouse environment should ne based on surrogate keys, not natural keys.



For optimisation, every surrogate key has be an numeric value.

None of the keys should be:

- Smart where you can tell something about the record just by looking at the key (except date key value).
- Composed of natural keys glued together.
- Implemented as multiple parallel joins between the dimension table and the fact table, so-called double or triple barreled joins.



Surrogate key

Surrogate key

- Uniquely identifies an instance of a dimension entity (i.e. a row)
- Usually a simple integer value
- Must be unique in the dimension table

Alternate key

- Identifies an entity in the operational source system
- Often a *business* key (e.g. a product code or customer ID) or a *natural* key (e.g. a datetime value in a time dimension)
- Can be duplicated in the dimension table to represent the same entity at different points in time

CustomerKey	CustomerAltKey	Name	Email	Street	City	PostalCode	CountryRegion
123	I-543*	Navin Jones	navin1@contoso.com	1 Main St.	Seattle	90000	United States
124	R-589	Mary Smith	mary2@contoso.com	234 190th Ave	Buffalo	50001	United States
125	I-321	Antoine Dubois	antoine1@contoso.com	2 Rue Jolie	Paris	20098	France
126	I-543*	Navin Jones	navin1@contoso.com	24 125th Ave.	New York	50000	United States
…	…	…	…	…	…	…	…

* *This customer moved from Seattle to New York, so a new record with the same alternate key but a new surrogate key was added.*

Dimension Table Core Concepts

Date & Time dimension

Every fact should have at least one date/or time dimension associated with.

Column Name	Data Type	Allow Nulls
TimeKey	int	☐
FullDateAlternateKey	datetime	☑
DayNumberOfWeek	tinyint	☑
EnglishDayNameOfWeek	nvarchar(10)	☑
SpanishDayNameOfW…	nvarchar(10)	☑
FrenchDayNameOfWeek	nvarchar(10)	☑
DayNumberOfMonth	tinyint	☑
DayNumberOfYear	smallint	☑
WeekNumberOfYear	tinyint	☑
EnglishMonthName	nvarchar(10)	☑
SpanishMonthName	nvarchar(10)	☑
FrenchMonthName	nvarchar(10)	☑
MonthNumberOfYear	tinyint	☑
CalendarQuarter	tinyint	☑
CalendarYear	char(4)	☑
CalendarSemester	tinyint	☑
FiscalQuarter	tinyint	☑
FiscalYear	char(4)	☑
FiscalSemester	tinyint	☑
		☐

Dimension Table Core Concepts

Slowing Changing Dimension

Type 1 : Overwrite

This methodology overwrites old with new data, and therefore does not track historical data.

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State
123	ABC	Acme Supply Co	CA

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State
123	ABC	Acme Supply Co	IL

The disadvantage of the Type 1 method is that there is no history in the data warehouse. It has the advantage however that it's easy to maintain.

http://www.kimballgroup.com/2005/03/slowly-changing-dimensions-are-not-always-as-easy-as-1-2-3/

Dimension Table Core Concepts

Slowing Changing Dimension

Type 2 : Add new row

This method tracks historical data by creating multiple records for a given natural key in the dimensional tables with separate surrogate keys and/or different version numbers.

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State	Version.
123	ABC	Acme Supply Co	CA	0
124	ABC	Acme Supply Co	IL	1

Supplier_Key	Supplier_Code	Supplier_Name	Supplier_State	Start_Date	End_Date
123	ABC	Acme Supply Co	CA	01-Jan-2000	21-Dec-2004
124	ABC	Acme Supply Co	IL	22-Dec-2004	*NULL*

upplier_Key	Supplier_Code	Supplier_Name	Supplier_State	Effective_Date	Current_Flag
123	ABC	Acme Supply Co	CA	01-Jan-2000	N
124	ABC	Acme Supply Co	IL	22-Dec-2004	*Y*

If there are retroactive changes made to the contents of the dimension, or if new attributes are added to the dimension (for example a Sales_Rep column) which have different effective dates from those already defined, then this can result in the existing transactions needing to be updated to reflect the new situation. This can be an expensive database operation, so Type 2 SCDs are not a good choice if the dimensional model is subject to change

http://www.kimballgroup.com/2005/03/slowly-changing-dimensions-are-not-always-as-easy-as-1-2-3/

Dimension Table Core Concepts

Slowing Changing Dimension

Type 3 : add new attribute

This method tracks changes using separate columns and preserves limited history.

Supplier_Key	Supplier_Code	Supplier_Name	Original_Supplier_State	Effective_Date	Current_Supplier_State
123	ABC	Acme Supply Co	CA	22-Dec-2004	IL

The Type 3 preserves limited history as it is limited to the number of columns designated for storing historical data.

http://www.kimballgroup.com/2005/03/slowly-changing-dimensions-are-not-always-as-easy-as-1-2-3/

Dimension Table Core Concepts

Slowing Changing Dimension

Type 0: No changes allowed

DateKey	DateAltKey	Day	Month	Year
20230101	01-01-2023	Sunday	January	2023

Type 1: Changes made inline in dimension row

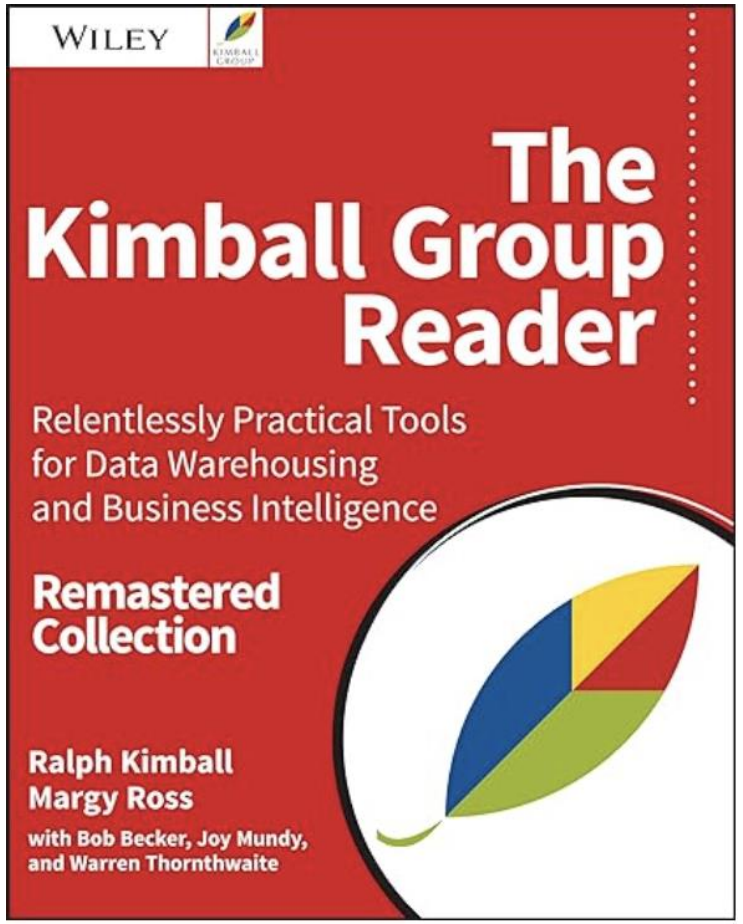
StoreKey	StoreAltKey	StoreName
123	EH199J	~~High Street Store~~ Town Central Store

Type 2: Changes result in a new version of the dimension entity (a new row)

CustomerKey	CustomerAltKey	Name	Address	City	DateFrom	DateTo	IsCurrent
1211	jo@contoso.com	Jo Smith	9999 Main St	Seattle	20190101	20230105	False
2996	jo@contoso.com	Jo Smith	1234 9th Ave	Boston	20230106		True

The Kimball Group Reader





https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/books/kimball-reader/